KAIF KHAN

Full-Stack Software Engineer · Systems & Web Infrastructure
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Software engineer with 4+ years of experience building high-reliability full-stack systems, transactional database architectures, and developer infrastructure. Proven track record implementing append-only financial ledgers, zero-downtime database migration pipelines, and token-efficient AI agent infrastructure. Strong background spanning backend systems correctness and accessible, high-performance UI engineering.

EXPERIENCE

Carpe Diam / Univ Diam — Product Engineer · Mumbai · Aug 2025 – Present

UnivDiam: made-to-order B2B jewelry manufacturing & ordering platform (React, Node.js, Express, MySQL, Socket.IO, Turborepo)

- Core engineer on the flagship B2B platform; scaled to 20 retail enterprise accounts and 240+ orders, growing monthly volume 4.5x (14 to 63 orders) and raising customer self-serve ordering adoption from 21% to 60%.
- Hardened platform security: resolved critical Broken Object Level Authorization (BOLA/IDOR) flaws by binding authorization checks to cryptographically signed JWT sessions, sanitized XSS render paths, and added token revocation checks to WebSocket handshakes.
- Engineered real-time order messaging via WebSockets with optimistic client-side updates, server acknowledgment reconciliation, and per-user read receipts; built an analytics engine computing funnel velocity directly from immutable activity logs.

Level Up: factory QC inspection platform (React, Express, PostgreSQL, S3, Docker, Capacitor/Android)

- Replaced unsafe runtime schema synchronization with 19 deterministic, sequenced migrations integrated into CI/CD against ephemeral PostgreSQL instances, eliminating 3 hidden schema regressions before beta launch.
- Re-architected photo ingestion from 5 MB base64 JSON payloads to presigned S3 direct uploads with magic-byte validation, reducing API body limits to 100 KB and optimizing query latency with compound indexes.
- Built an offline inspection workstation on Android (Capacitor) with an indexed local outbox and sync queue; refactored 1,350+ line view into modular domain components (down to 389 lines) with table virtualization and WCAG keyboard accessibility.

CAD Portal: internal CAD workflow & payout platform (Next.js, PostgreSQL, Drizzle ORM)

- Architected an authoritative finite state machine pipeline enforcing status transitions, stage cycle times, role-scoped task queues, and dynamic rate cards automating monthly designer compensation.
- Validated state transition invariants against 100% of historical production tasks (64/64) to resolve data anomalies prior to zero-downtime cutover; mentored 2 engineering interns on system design and S3 upload architecture.

VCBay Solutions — UX/UI Designer & Web Designer — Frontend Development Focus · Remote · Jun 2022 – Jul 2025

- Collaborated across cross-functional engineering and product teams to build user-facing React interfaces for Zefyron GmbH (enterprise B2B SaaS), streamlining workflows across 4 stakeholder roles.
- Created and maintained shared design systems bridging Figma tokens to frontend components, ensuring interface consistency and supporting efficient developer handoffs.
- Built and launched production corporate websites for 5+ client brands using Webflow and custom code, integrating dynamic CMS structures and responsive interactions.

SAMISON Corporate — Software Engineering Intern · Remote · Jun 2021 – Sep 2021

- Developed modular React UI components and data-fetching pipelines, collaborating with senior engineers to optimize client-side bundle size, rendering performance, and cross-browser compatibility.

PROJECTS

ContextQuant (ai-gzip) — Token Optimization & MCP Gateway · TypeScript, Bun, WebAssembly, C · github.com/khankaif/ContextQuant

- Architected a high-throughput proxy gateway for Model Context Protocol (MCP) servers that reduces LLM tool-calling token consumption by 85%–95% (e.g., 37.9k down to 5.5k tokens on live GitHub MCP responses) without silent data loss.
- Implemented dynamic schema projection, invariant hoisting (factoring out redundant fields across record sets), and deterministic receipting with on-demand hydration, validated across 10,000+ records using real tokenizers (`o200k`).

Commodity ERP & Double-Entry Financial Ledger · Next.js, PostgreSQL, TypeScript · In Production

- Designed and built an ERP and double-entry financial ledger for a diamond trading firm, managing stock, consignment memos, receivables, payables, and live P&L in production, replacing fragmented spreadsheets.
- Enforced transactional data integrity via append-only ledgers with triggers blocking mutations (requiring reversal entries) and pessimistic row-locking to eliminate inventory oversell race conditions; migrated and reconciled 13,000+ historical records across ₹1.59 crore with negligible ₹0.10 drift.

SKILLS

Languages: TypeScript, JavaScript, SQL, C, HTML/CSS
Backend & Systems: Node.js, Express, Next.js, PostgreSQL, MySQL, Redis, Drizzle ORM, Sequelize, REST APIs, WebSockets, Concurrency Control
Frontend & Mobile: React, Next.js, TanStack Query, Tailwind CSS, Vite, Capacitor (Android), WCAG Accessibility
Infrastructure & Tools: Docker, Nginx, GitHub Actions, AWS S3, Turborepo, Sentry, Linux/Bash, Model Context Protocol (MCP)

EDUCATION

B.E. Computer Engineering — Rizvi College of Engineering, University of Mumbai · 2018 – 2022